

Homework: Blockchain Tech

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TOPIC: UPORT: BLOCKCHAIN IDENTITY PROJECT

Introduction

uPort is a **decentralized digital identity platform** built on the **Ethereum blockchain**. It allows users to create and manage their own identities without relying on centralized authorities like governments or corporations. With uPort, individuals control their personal data and decide how and when to share it. This aligns with the concept of **self-sovereign identity (SSI)**, where users own their identity and verification happens through trusted blockchain transactions.

The platform was developed by **ConsenSys**, one of the leading blockchain development companies, and was designed to solve a major issue in today's digital world — the lack of trust and privacy in online identity management. Traditional systems rely on centralized databases, which are vulnerable to hacking, data leaks, and misuse. uPort offers a blockchain-based alternative that is more transparent and user-centric.

How uPort Works ?

uPort uses **smart contracts** on Ethereum to register, manage, and verify identities. Each user has:

1. A **uPort ID** (a unique blockchain address)
2. A **mobile app** to store and manage credentials
3. A **recovery system** in case of lost devices

When a user registers, a smart contract is created on the blockchain that links their public key to their identity. The user can then use this identity to log in to websites, sign documents, or share verified credentials — without a password or centralized login system. Verification happens through **cryptographic signatures**, ensuring security and authenticity.

uPort: Strengths and Weaknesses

Strengths

1. **User Control** – Users own their identity and decide what information to share.
2. **Privacy** – No central server stores data, reducing the risk of leaks or misuse.
3. **Security** – Built on Ethereum, using cryptography and blockchain for protection.
4. **Transparency** – All identity actions can be verified on the blockchain.
5. **Interoperability** – Works with global standards like DIDs and Verifiable Credentials.
6. **Mobile Friendly** – The uPort app is easy to use for identity management anywhere.
7. **Decentralized System** – No single authority controls the data or network.

Weaknesses

1. **High Transaction Costs** – Ethereum gas fees can make uPort expensive to use.
2. **Slow Transactions** – Network congestion may delay verification or registration.
3. **Limited Legal Status** – Not officially recognized by most governments.
4. **Recovery Difficulty** – If users lose their keys or trusted contacts, recovery is hard.
5. **Public Data Visibility** – Some activity details are visible on the blockchain.
6. **Low Adoption** – Few organizations and users know how to use decentralized ID.